FEDERICO TIBERGA

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EDUCATION

London Business School, London, United Kingdom — Spring 2029
PhD in Management Science and Operations
Relevant Coursework: Stochastic Processes, Econometrics, Advanced Optimization

Cornell Tech, New York, NY — May 2023
Master of Engineering in Operations Research and Information Engineering | GPA: 4.0
Relevant Coursework: Pricing Analytics and Revenue Management, Applied Machine Learning, Optimization Methods
Independent Research – Greedy Algorithms for Assortment Optimization

Politecnico di Torino, Torino, Italy. — Jul 2022
Bachelor's in Mathematics for Engineering – Young Talents Program | GPA: 3.8
Relevant Coursework: Probability and Statistics, Database, Linear Algebra, Multivariable Calculus, Partial Differential Equations.

Collegio Carlo Alberto, Torino, Italy. — Jul 2022
Diploma in Economics – Allievi Honors Program | GPA: 3.87
Relevant Coursework: Microeconomic Theory, Models and Algorithms, Optimization for Economics.

EXPERIENCE

Efficio, New York, NY
Consultant I — Jul 2024 – Sep 2024
Business Analyst — Aug 2023 – Jul 2024

- Devised and executed cost-saving initiatives by optimizing cloud services reservations and saving plans, achieving up to 30% reduction in annual expenses for the client, all while preserving peak performance and reliability.
- Developed and implemented an optimization model for Azure instances rightsizing using Azure Monitor, enhancing performance and reliability while achieving $1M+ annualized savings for multiple clients.
- Created a price anomaly tracking system that proactively identifies pricing discrepancies and provides real-time insights to clients, enabling them to promptly address anomalies and optimize spending.
- Collaborated with cross-functional teams to gather and analyze business requirements, leveraging vast datasets to translate them into actionable insights and data-driven solutions.

PROJECTS

A Machine Learning Model to Predict Tennis Matches (Python, MySQL) — Fall 2022 - Current

- Developed a custom variation of the Glicko-2 rating system for predicting professional tennis outcomes.
- Gathered and cleaned a comprehensive ATP and WTA Tour datasets, focusing on players' skill level, form, fatigue, head-to-head, and performance against common adversaries and players with similar playstyle as the opponent
- Implemented an LSTM machine learning algorithm. Extensive research was conducted, including feature elimination to mitigate overfitting, hyperparameter optimization, and uncertainty reduction, all contributing to the model's effectiveness.
- Achieved 70% prediction accuracy on the dataset of over 400k games, and an average ROI of 7% per match played on the test set, showcasing the practical application and success of the machine learning model.

Greedy Algorithms for Assortment Optimization (Python) — Spring 2023

- Investigated assortment optimization in e-retail and choice modeling techniques.
- Analyzed greedy algorithms' performance in optimization, emphasizing speed and simplicity.
- Conducted extensive experimental analysis with various price and attraction value distributions.
- Utilized probabilistic analysis to evaluate real-world performance of the algorithm.

SKILLS

Coding Languages:	Python (NumPy, SciPy, Pandas, PyTorch, scikit-learn), SQL, MATLAB, Julia, C++
Languages:	Italian (Native), English (IELTS 8.0 – C1), French (DALF C1)

CERTIFICATIONS AND AWARDS

Italian Mathematical Olympiads — May 2019

- Silver medal, individual competition
- 2nd place, team competition